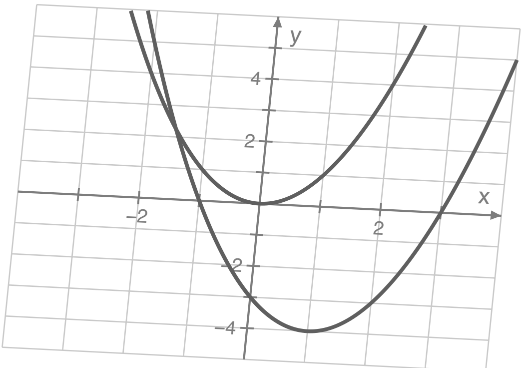
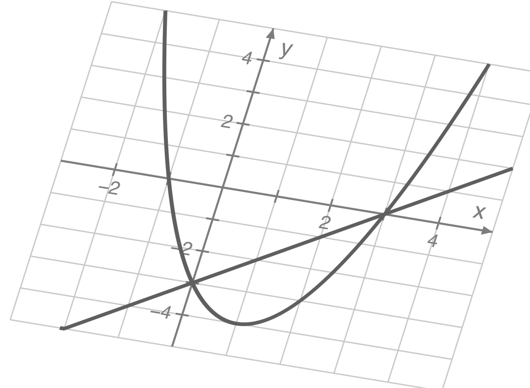


$$f(x) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$x = -\frac{b}{2a}$$

$x^2 - 2x - 3 = 0$
$(x+1)(x-3) = 0$
$x = -1 \quad x = 3$

$x^2 - 2x = 3$
$(x-1)^2 = 4$
$x = 1 \pm 2$

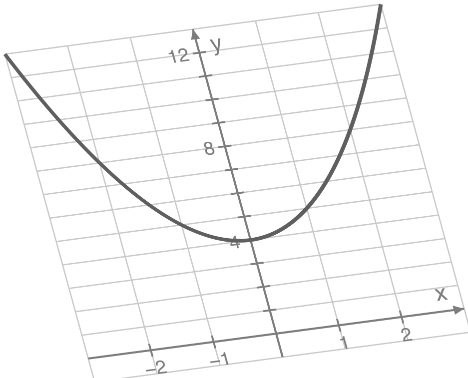
$$x^2 = k \square \quad x = \pm\sqrt{k}$$

$$f(x) = a(x-p)(x-q)$$

Algebra 2

Unit 2: Quadratic Functions

Shepherd



$$i = \sqrt{-1} \quad i^2 = -1$$

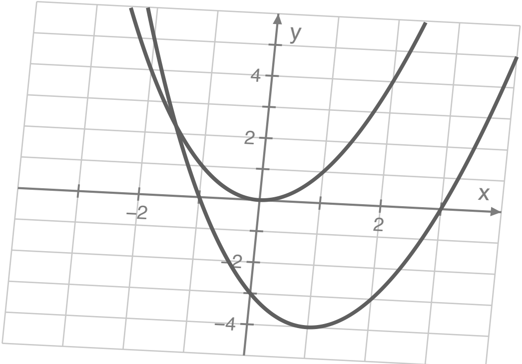
$x^2 + 2x + 5 = 0$
$(x+1)^2 = -4$
$x = -1 \pm 2i$

$$(a + bi)(a - bi) = a^2 + b^2$$

$b^2 - 4ac > 0$
$b^2 - 4ac = 0$
$b^2 - 4ac < 0$

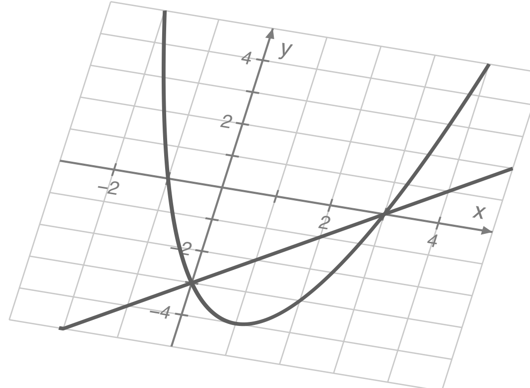
$$h(t) = -16t^2 + 32t + 48$$

$$A(x) = x(40 - 2x)$$



$$f(x) = a(x-h)^2 + k$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



$$x = -\frac{b}{2a}$$

$$x^2 - 2x - 3 = 0$$
$$(x+1)(x-3) = 0$$
$$x = -1 \quad x = 3$$

$$x^2 = k \square \quad x = \pm\sqrt{k}$$

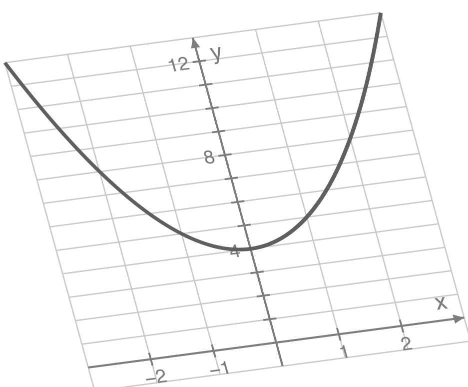
$$x^2 - 2x = 3$$
$$(x-1)^2 = 4$$
$$x = 1 \pm 2$$

$$f(x) = a(x-p)(x-q)$$

Algebra 2

Unit 2: Quadratic Functions

Shepherd



$$i = \sqrt{-1} \quad i^2 = -1$$

$$x^2 + 2x + 5 = 0$$
$$(x+1)^2 = -4$$
$$x = -1 \pm 2i$$

$$(a+bi)(a-bi) = a^2 + b^2$$

$$b^2 - 4ac > 0$$
$$b^2 - 4ac = 0$$
$$b^2 - 4ac < 0$$

$$h(t) = -16t^2 + 32t + 48$$

$$A(x) = x(40 - 2x)$$